Computational Statistics
Unit 6: Bootstrapping

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1. Introduction to Bootstrap and Resampling methods

- The main issue with Statistics is that quantifying errors of estimators based on real-data is very hard
- Much of the usual error terms are obtained via asymptotic theoretical results which work under different assumptions that may or not hold
- Bootstrap and resampling methods in general allow us to use simulation-based method to measure the accuracy of our estimates without the need of having the support of theoretical assumptions
- Bootstrapping was introduced by Bradley Efron in 1979 and completely modernised the field
- Brad Efron is one of the heroes Modern Statistics

Motivation: Introduction to Statistical Inference

To understand why simulations method are needed within Statistics is it good to analyse what are the problems of classic Statistical theory. Let's start by remembering some basics of Statistical Inference

- Consider i.i.d. data $X_1, \ldots, X_n$ from a population.
- Suppose that $X_i \sim \mathcal{N}(\mu, \sigma^2)$ with μ unknown but σ^2 is known
- We want to estimate μ and to obtain a 99% confidence interval.
- the usual estimate is
$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$
- For the confidence interval it would be good to have the distribution of $\hat{\mu} - \mu$, which is possible
- $\hat{\mu} - \mu \sim \mathcal{N}(0, \frac{\sigma^2}{n})$ or rather $\frac{\sqrt{n}(\hat{\mu}-\mu)}{\sigma} \sim \mathcal{N}(0,1)$
- Set $\alpha = 0.001$ and let $a < b$ be such that $\mathbb{P}(\mathcal{N}(0,1) \in [a,b]) = (1-\alpha)$. Set $I_{\alpha,n}$ by
$$I_\alpha = \left[\hat{\mu} + a\frac{\sigma}{\sqrt{n}}, \hat{\mu} + b\frac{\sigma}{\sqrt{n}}\right]$$
- Note that $I_{\alpha,n}$ is a random interval

Motivation: Introduction to Statistical Inference

Proposition

$$\mathbb{P}(\mu \in I_\alpha) = 1 - \alpha$$

Proof: exercise

Motivation: Introduction to Statistical Inference

- Let's drop the assumption that we know the variance σ^2.
- We still have $\frac{\sqrt{n}(\hat{\mu}-\mu)}{\sigma} \sim \mathcal{N}(0,1)$ but we need to estimate σ^2
- The usual estimator is

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})^2$$

- what is the distribution of $\frac{\sqrt{n}(\hat{\mu}-\mu)}{\hat{\sigma}}$?
- Answer: T-student with $n-1$ degree of freedom

$$\frac{\sqrt{n}(\hat{\mu}-\mu)}{\hat{\sigma}} \sim T_{n-1}$$

- We can now find an interval $[a,b]$ such that $\mathbb{P}(T_{n-1} \in [a,b]) = 1-\alpha$ and then we have that

$$I_{\alpha,n} = \left[\hat{\mu} + a\frac{\hat{\sigma}}{\sqrt{n}}, \hat{\mu} + b\frac{\hat{\sigma}}{\sqrt{n}}\right]$$

  is a $1-\alpha$ confidence interval for μ.

Motivation: Introduction to Statistical Inference

- Let's go even further and let's drop the assumption that $X_i \sim \mathcal{N}(\mu, \sigma^2)$ and let's just assume that X_i has some distribution with mean μ and variace σ
- There are plenty of paths to follow here. Let's just not assume anything about the distribution that produced our data.
- It we have few data points, there is nothing we can do, but if we have a large amount of data points we can still do Statistics
- By the Central Limit Theorem we have

$$\sqrt{n}\frac{\hat{\mu} - \mu}{\sigma} \xrightarrow{\mathcal{D}} \mathcal{N}(0, 1)$$

  as $n \to \infty$
- But we still need to estimate $\hat{\sigma}$ using the usual estimate

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{\mu})$$

Motivation: Introduction to Statistical Inference

- We still need some help from Probability theory
- Exercise: Use the law of Large Numbers to show that

$$\hat{\sigma} \to \sigma$$

  as $n \to \infty$
- There is a key result here: Slutsky's theorem

Proposition

Suppose that $X_n \xrightarrow{\mathcal{D}} X$ and that $Y_n \to y$, where y is constant. then

$$Y_n X_n \xrightarrow{\mathcal{D}} yX$$

- Then

$$\sqrt{n}\frac{\hat{\mu}-\mu}{\hat{\sigma}} = \left(\frac{\sigma}{\hat{\sigma}}\right)\left(\sqrt{n}\frac{\hat{\mu}-\mu}{\sigma}\right) \xrightarrow{\mathcal{D}} \mathcal{N}(0,1)$$

- Consider $a < b$ such that $\mathbb{P}(\mathcal{N} \in [a, b]) = 1 - \alpha$, then

$$I_{\alpha,n} = \left[\hat{\mu} + a\frac{\sqrt{n}}{\hat{\sigma}}, \hat{\mu} + b\frac{\sqrt{n}}{\hat{\sigma}}\right]$$

  is an asymptotic $(1-\alpha)$ confidence interval for μ

Motivation: Introduction to Statistical Inference

- We have the following

Proposition

$$\lim_{n \to \infty} \mathbb{P}\left(\mu \in I_{\alpha,n}\right) = 1 - \alpha.$$

- So far, Statistics is working very well.

Motivation: Introduction to Statistical Inference

What if we are interested on other statistics?

- If our statistics have the form $\frac{1}{n}\sum_{i=1}^{n} h(X_i)$ we can easily compute mean and variance and be happy.
- e.g. $X_i = (H_i, W_i)$, then $h(X_i) = W_i/H_i^2$
- things are difficult when the quantity we want to estimate is not a sum, for example, **the median**

Motivation: Statistical Inference for the median

- Let $(X_i)_{i=1}^n$ be i.i.d. data in $\mathbb{R}$
- Let's just assume that the distribution function is continuous
- A reasonable estimator for the median is given by $X_{\lceil n/2 \rceil : n}$, the value that is in position $\lceil n/2 \rceil$ when we sort the data in increasing order
- What can we do in this case?
- Let's do a bit of simulation to get some intuition.

Motivation: Statistical Inference for the median

1. We still have no idea what to do with our estimator $X_{\lceil n/2 \rceil : n}$
2. but something can be done...

Motivation: Statistical Inference for the median

Let $U_1, \ldots, U_n \overset{iid}{\sim} \mathcal{U}(0,1)$.

Theorem

$$2\sqrt{n}(U_{\lceil n/2 \rceil} - 1/2) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2 = 1)$$

The proof is standard and can be found in several textbooks

Delta Method

The delta method is a powerful method that allow us to prove asymptotic normality and it is very effective in Data Analysis

Theorem (Delta Method)

Suppose that $\sqrt{n}(S_n - s) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2)$, *and let* $g : \mathbb{R} \to \mathbb{R}$ *be such that* $g'(s)$ *exists and it is non-zero. Then*

$$\sqrt{n}(g(S_n) - g(s)) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2[g'(s)]^2)$$

There is also a multivariate version if you want to investigate further

Motivation: Statistical Inference for the median

Let $U_1, \ldots, U_n \overset{iid}{\sim} \mathcal{U}(0, 1)$. Let $M_n = U_{\lceil n/2 \rceil}$

Theorem

$$\sqrt{n}(M_n - 1/2) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2 = 1/4)$$

We can now combine with the Delta Method. We now that for a given continuous distribution F we have that $F^{-1}(M_n)$ has the distribution of the empirical median of n independent samples from F.

Theorem

Let F be a continuous distribution with $F(m) = 1/2$ and $F'(m) \neq 0$. Consider data $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F$, then

$$\sqrt{n}(X_{\lceil n/2 \rceil} - m) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2 = 1/(2F'(m))^2)$$

Motivation: Statistical Inference for the median

- Under certain conditions we have that

$$\sqrt{n}(X_{\lceil n/2 \rceil} - 1/2) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2 = 1/(2F'(m))^2)$$

- so we should be ready to provide confidence intervals for the Median
- but we would need to compute $F'(m)$, which of course we don't know
- and estimation of density functions is extremely hard
- so all our effort didn't paid off
- unless we were able to estimate $\mathbb{V}ar(\sqrt{n}(X_{\lceil n/2 \rceil} - 1/2))$ in which case we can approximate $1/(2F'(m))^2$ by such an estimate

Motivation: Statistical Inference for the median

The main idea of Bootstrap is the following.

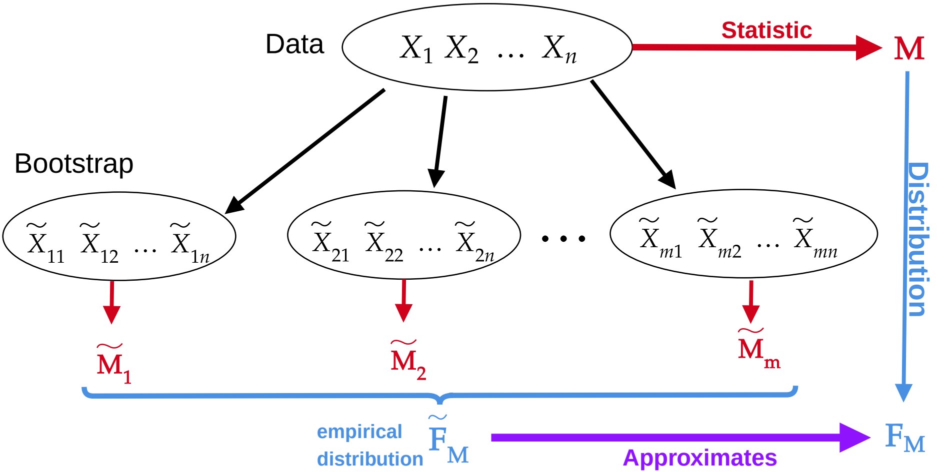
- Given **observed** data $X_1, \dots, X_n$, we can define the so-called empirical distribution P_n which put mass $1/n$ over each observation X_i
- And we can sample from it! A random variable $\tilde{X} \sim P_n$ can be sampled by choosing a random index I and then taking the value X_I
- Let $\tilde{X}_1, \dots, \tilde{X}_n$ be i.i.d. samples from P_n. We can compute the median of such samples, say $\tilde{M} = \tilde{X}_{\lceil n/2 \rceil}$
- Consider $\tilde{M}_1, \dots, \tilde{M}_m$ as m independent bootstrap copies of $\tilde{M}$ (each of them computed from independent data)
- We can use them to estimate the expectation of $X_{\lceil n/2 \rceil}$ and its variance.
- Magically, everything works fine! as long as the set of observed data is large enough.
- Let's see the code...

Abstract Bootstrap idea

The idea is simply:

1. a statistic is a function of the data.
2. each data set gives us only one value for the statistic
3. bootstrap creates 'fake datasets'
4. now we have several copies of the statistic and gather info about **its distribution**
5. it works pretty well, even if we cannot provide any theory about estimator (more or less)

Abstract Bootstrap idea



2. Efron's Bootstrap

Efron's Bootstrap

Efron's idea to construct bootstrapped data sets is to resample directly from $\{X_1, \ldots, X_n\}$

Efron Sampling Method

1. Consider data $D = (X_1, \ldots, X_n)$
2. For $i = 1$ to m do:
   2.1 Sample $I_1, \ldots, I_n$ as i.i.d. random variables with uniform in $\{1, \ldots, n\}$.
   2.2 Set $\widetilde{X}_{ij} = X_{I_j}$ for all $j \in \{1, \ldots, n\}$
   2.3 Set $\widetilde{D}_i = (\widetilde{X}_{i1}, \ldots, \widetilde{X}_{in})$
3. return $(\widetilde{D}_1, \ldots, \widetilde{D}_m)$

In the previous example: our statistic is the median of the data, i.e. $t(D) = X_{\lceil n/2 \rceil}$. We can then estimate $\mathbb{V}ar(t(D))$ by $\widetilde{\sigma}^2$

- $\widetilde{\mu} = \frac{1}{m} \sum_{i=1}^{m} t(\widetilde{D}_i)$
- $\widetilde{\sigma}^2 = \frac{1}{m-1} \sum_{i=1}^{m} \left(t(\widetilde{D}_i) - \widetilde{\mu} \right)^2$

Then, we can construct an approximate confidence interval for the median by using that by using the quantiles of a $\mathcal{N}(\widetilde{\mu}, \widetilde{\sigma}^2)$

Efron's Bootstrap

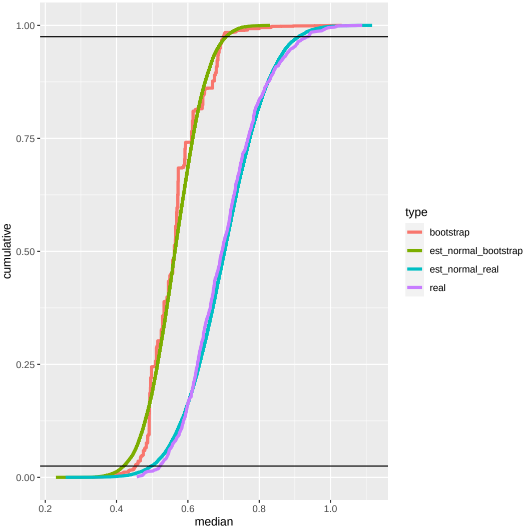
We can go further and forget about theory. We do not need to use a normal approximation to build a confidence interval.

A bootstrap confidence interval for the median

1. Let $B = (\widetilde{M}_1, \ldots, \widetilde{M}_m)$ be m bootstrap estimators for the median based on a real-valued data $D = (X_1, \ldots, X_n)$
2. Let $\widetilde{q}_{\alpha/2}$ and $\widetilde{q}_{1-\alpha/2}$ be the α and $1-\alpha$ quantile of the sample B
3. Then we have a confidence interval.. and the simulations show it seems to work well

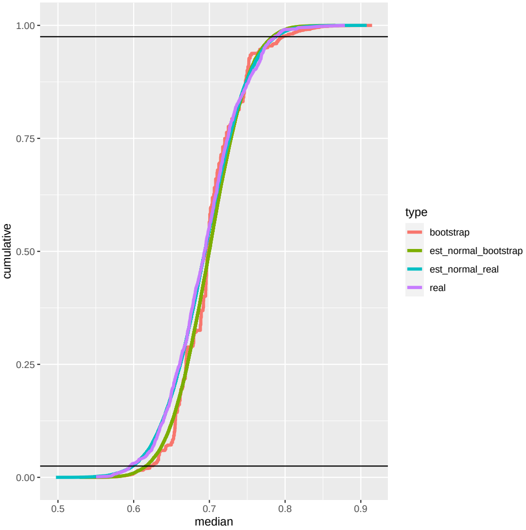
Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{100} \overset{iid}{\sim} \text{Exp}(1)$



Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{400} \overset{iid}{\sim} \text{Exp}(1)$



Efron's Bootstrap for the median

Bootstrapping is very helpful even if classic statistical theory "fails". Let's see the following simple example:

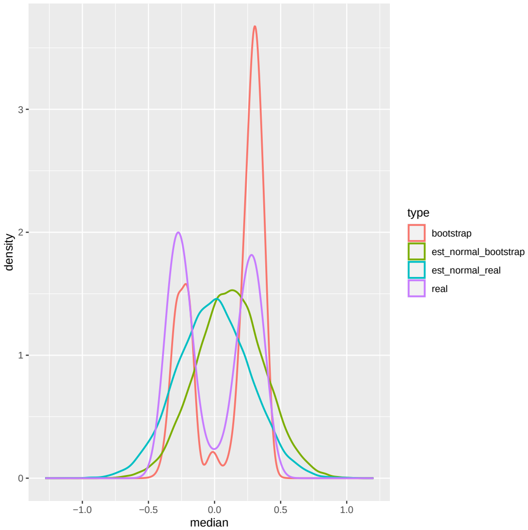
1. Consider

$$f(x) = \frac{3}{2}x^2; \quad x \in [-1, 1]$$

2. clearly the median of f is 0, and $f(0) = 0$. Therefore, the convergence to normal of the median estimator is not guaranteed. Indeed, we can show it is not normal.
3. But bootstrap will do fine!

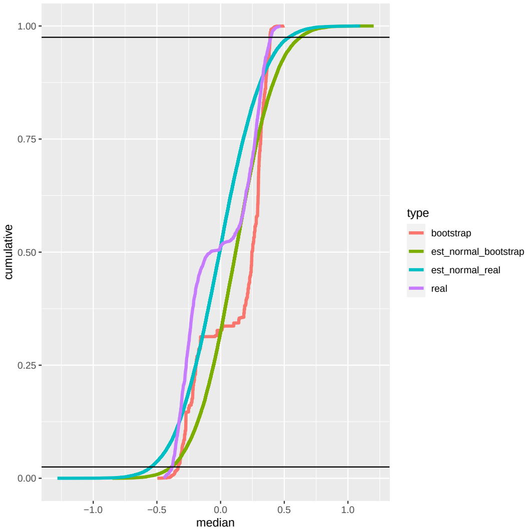
Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{1000} \overset{iid}{\sim} (3/2)x^2 \mathbb{1}_{\{-1 \leq x \leq 1\}}$



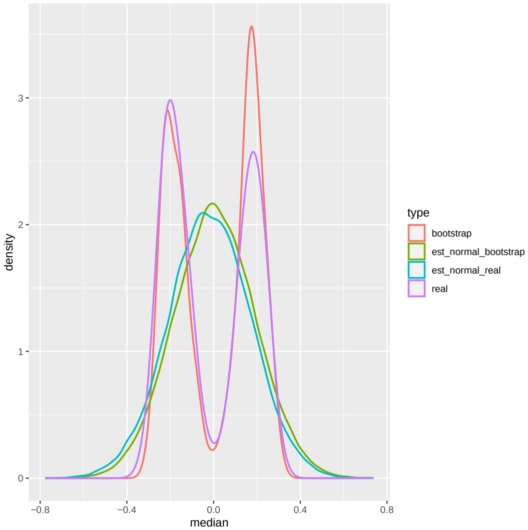
Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{1000} \overset{iid}{\sim} (3/2)x^2 \mathbb{1}_{\{-1 \leq x \leq 1\}}$



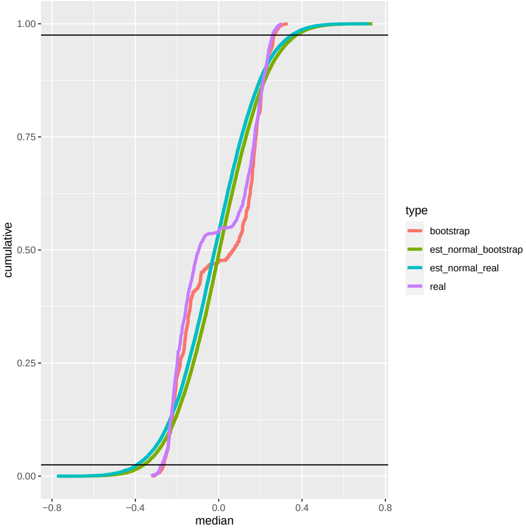
Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{\mathbf{10000}} \overset{iid}{\sim} (3/2)x^2 \mathbb{1}_{\{-1 \le x \le 1\}}$



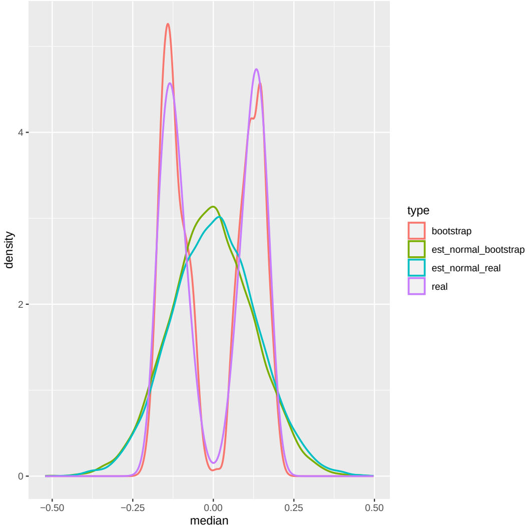
Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{10000} \overset{iid}{\sim} (3/2)x^2 \mathbb{1}_{\{-1 \leq x \leq 1\}}$



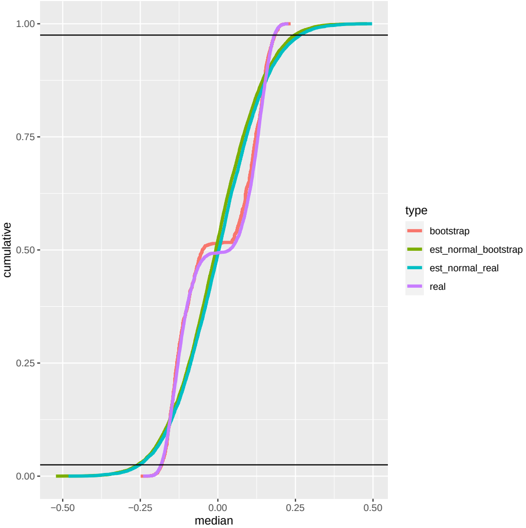
Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{\mathbf{100000}} \overset{iid}{\sim} (3/2)x^2 \mathbb{1}_{\{-1 \leq x \leq 1\}}$



Efron's Bootstrap for the median

Data: $X_1, \ldots, X_{\mathbf{100000}} \overset{iid}{\sim} (3/2)x^2 \mathbb{1}_{\{-1 \leq x \leq 1\}}$



Efron's Bootstrap for the median

- In our example everything seems to work well...
- but it is quite data-expensive
- and it takes a lot of time to compute

General construction of confidence intervals

- Consider i.i.d. data $X_1, X_2, \ldots, X_n$
- Consider a statistic $S(X_1, \ldots, X_n) \in \mathbb{R}$
- Using Efron's Bootstrap obtain $S_B = (\tilde{S}_1, \ldots, \tilde{S}_M)$, each $\tilde{S}_i$ is calculated with a bootstrap sample
- Sort S_B in increasing order and choose (a, b) as the $\alpha/2$ and $1 - \alpha/2$ quantile, respectively
- (a, b) is the $1 - \alpha$ bootstrap confidence interval for $S(X_1, \ldots, X_n)$.

When does Bootstrap work?

The general rule for Bootstrap is

Do not mess with tails

meaning that estimators that depend too much on the extreme values of the distribution will likely fail.

Some examples

1. mean, variances, and moments in general work fine: be careful because large moments like $\mathbb{E}(X^8)$ start depending too much on extreme values, so more data will be needed
2. Median and quantiles are work fine: be careful will quantiles like 1% as they depend a lot on the extreme values, but with a lot of data, 1% is still a lot of data.
3. Extreme values such as the maximum or minimum value of a distribution are **bad**, and bootstrap do not work there.

3. Other type of Bootstrap

Hypothesis Testing

Remember than in Hypothesis Testing we want to find rejection regions.

Usually, we want to construct a test-statistic T that has to **very different** behaviours under the null and the alternative

So we construct a rejection regions by studying the distribution of T under the null hypothesis

Usually, this requires some theory

Bootstrap for Hypothesis Testing

We may want to use **Bootstrap** to find rejection regions avoiding using theory

The difficulty is that under the alternative, **Efron's Bootstrap** does not sample data assuming the null distribution. Actually, it will just imitate the behaviour of the distributions under the same alternative

So we need new forms of Bootstrapping or Resampling our data, that gives us 'data' under the null

Unfortunately, we will need some theory, since the null hypothesis carries information that has to be incorporated in our construction

In the end, we may need to **construct** and **prove** that our **bootstrap** method work, but there are some common ideas that are useful to know

Bootstrap by permutations: two-sample problems

Bootstrap by Permutations is very common in settings were we are comparing distributions.

Consider data in $\mathbb{R}^d$ given by $X_1, \ldots, X_n \overset{i.i.d.}{\sim} F$ and $Y_1, \ldots, Y_m \overset{i.i.d.}{\sim} G$, and that both samples are independent. For short, denote by $\mathbf{X}$ and $\mathbf{Y}$ the set of data $(X_i)_{i=1}^n$ and $(Y_j)_{j=1}^m$ respectively.

We want to test if $H_0 : F = G$, against the alternative $H_1 : F \neq G$. This is called the **two-sample problem**.

Depending on the information we have about F and G we may be able to use theory to build a rejection region

Permutation tests

Permutation Tests:

1. Let $t(\mathbf{X}, \mathbf{Y})$ be a test-statistic in $\mathbb{R}$
2. We want $t(\mathbf{X}, \mathbf{Y}) \approx 0$ under the null
3. and $|t(\mathbf{X}, \mathbf{Y})| \gg 0$ under the alternative.
4. The typical example is

$$t(\mathbf{X}, \mathbf{Y}) = \frac{1}{n} \sum_{i=1}^{n} \omega(X_i) - \frac{1}{m} \sum_{i=1}^{m} \omega(Y_i)$$

5. To build a rejection region we 'fake' data from the null

Permutation tests. Cont

6. Consider $\mathbf{Z} = (X_1, \ldots, X_n, Y_1, \ldots, Y_m)$

7. Let $\sigma_1, \ldots, \sigma_M$ be i.i.d. random permutations of $\{1, 2, \ldots, n+m\}$

8. Denote by $\mathbf{Z}_{\sigma_i}$ the vector $\mathbf{Z}$ permuted according to σ_i

9. For each $i \in \{1, \ldots, M\}$ define $\mathbf{X}_{\sigma_i}$ as the first n elements of $\mathbf{Z}_{\sigma_i}$, and $\mathbf{Y}_{\sigma_i}$ the the last m elements of $\mathbf{Z}_{\sigma_i}$.

10. Let $t_i = t(\mathbf{X}_{\sigma_i}, \mathbf{Y}_{\sigma_i})$

11. Given α, define t^* as the largest possible value such that
$$\frac{1}{M}\sum_{i=1}^{M} \mathbb{1}\{|t(\mathbf{X}_{\sigma_i}, \mathbf{Y}_{\sigma_i})| \leq t^*\} \leq 1 - \alpha$$

12. Reject the null if $t(\mathbf{X}, \mathbf{Y}) \in R$, otherwise do not reject.

Permutation tests. Cont

To check that a test is valid, we want to control the type 1 error, that is

$$\mathbb{P}_{H_0}(\text{reject}) \leq \alpha.$$

Also, ideally we want that

$$\mathbb{P}_{H_1}(\text{reject}) \to 1$$

where the limit is taking as the number of data points increase. The later means that our test is powerful.

Permutation tests: analysis under the null

Under the null we have that all data points X_i and Y_j have the same distribution, so any permutation of them is indistinguishable.

Theorem

Let R be the rejection region as described in our algorithm. Then, under the null hypothesis we have that

$$\mathbb{P}_{H_0}(t(\mathbf{X}, \mathbf{Y}) \in R) \leq \alpha$$

Proof in whiteboard

Permutation tests: analysis under the alternative

Under the alternative all permutations of $(X_1, \ldots, X_n, Y_1, \ldots, Y_n)$ have the same distribution, and particularly $\mathbf{X}_{\sigma_\mathbf{i}}$ has the same distribution as $\mathbf{Y}_{\sigma_\mathbf{i}}$.

Theorem

Assume the alternative hypothesis and let R be the rejection region as described in our algorithm.
Moreover, suppose that $t(\mathbf{X}, \mathbf{Y})) \to c \neq 0$ as the number of data points grows to infinity. Then,

$$\mathbb{P}_{H_1}(t(\mathbf{X}, \mathbf{Y}) \in R) \to 1$$

as the number of data points grows.

The proof requires some advanced elements of probability, the so-called combinatorial CLT and conditional convergence in distribution, so we just give some ideas in the whiteboard.

Testing independence with permutation tests

- Consider now data in pairs $(X_i, Y_i)_{i=1}^n$ where each pair is sampled independently from a joint distribution $P_{X,Y}$.

- We want to test $H_0 : P_{X,Y} = P_X P_Y$ against the alternative $H_1 : P_{X,Y} \neq P_X P_Y$

- A typical test-statistic is a weighted covariance

$$t(\mathbf{X}, \mathbf{Y}) = \frac{1}{n} \sum_{i=1}^{n} f(X_i) g(Y_i) - \left(\frac{1}{n} \sum_{i=1}^{n} f(X_i) \right) \left(\frac{1}{n} \sum_{i=1}^{n} g(Y_i) \right)$$

Testing independence with permutation tests. Cont.

- We can use a permutation approach to build a rejection region. Let $\sigma_1, \ldots, \sigma_M$ be i.i.d. random permutations from $\{1, 2, \ldots, n\}$.
- Denote by $\mathbf{Y}_{\sigma_i}$ the vector $\mathbf{Y}$ permuted according to σ_i.
- Under the null hypothesis $\mathbf{X}, \mathbf{Y}$ has the same distribution as $\mathbf{X}, \mathbf{Y}_{\sigma_i}$
- let $t_i = t(\mathbf{X}, \mathbf{Y}_{\sigma_i})$
- Given α, define t^* as the largest possible value such that

$$\frac{1}{M}\sum_{i=1}^{M} \mathbb{1}\{|t(\mathbf{X}, \mathbf{Y}_{\sigma_i})| \leq t^*\} \leq 1 - \alpha$$

- Reject the null if $t(\mathbf{X}, \mathbf{Y}) \in R$, otherwise do not reject.

Testing independence with permutation tests. Cont.

The same argument used in the two-sample problem applies

Theorem

Let R be the rejection region as described in our algorithm. Then, under the null hypothesis we have that

$$\mathbb{P}_{H_0}(t(\mathbf{X}, \mathbf{Y}) \in R) \leq \alpha$$

Theorem

Assume the alternative hypothesis and let R be the rejection region as described in our algorithm.
Moreover, suppose that $t(\mathbf{X}, \mathbf{Y})) \to c \neq 0$ as the number of data points grows to infinity. Then,

$$\mathbb{P}_{H_1}(t(\mathbf{X}, \mathbf{Y}) \in R) \to 1$$

as the number of data points grows.

Some final Remarks

- We only had a look at basic bootstrapping ideas
- a perfect understanding of bootstrap require some non-trivial probability tools
- deriving bootstrap methods ad-hoc for the problem at hand requires some creativity
- there is no general strategy that we can follow

4. Extra

4. Extra

Bootstrap for Regression

Bootstrap for Regression

Suppose we have independent regression data $(X_i, Y_i)_{i=1}^n$ where $X_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$, and we have the relation

$$Y_i = \beta_0 + \sum_{j=1}^{d} \beta_j X_{ij} = \beta_0 + \beta^\intercal X_i$$

Depending on what we want to do there are several things we can do:

- If we already have formulas, we may use Efron's Bootstrap to get confidence intervals for coefficients, etc.
- For that, we need to consider our data as the full pair (X_i, Y_i)
- For example, if we denote $\beta = (\beta_0, \beta_1, \ldots, \beta_d)$, and the design matrix $\mathbf{X}$ by

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & & 1 \\ | & | & & | \\ X_1 & X_2 & \cdots & X_n \\ | & | & & | \end{pmatrix}$$

  Then the least squares estimator for β is

$$\hat{\beta} = (\mathbf{X}^\intercal \mathbf{X})^{-1} \mathbf{X}^\intercal y$$

- we can bootstrap such quantity

Bootstrap for Regression

- The problem with Efron's bootstrap is that if we resample data, the **bootstrap design matrix $\mathbf{X_B}$**, will have repeated columns, so $(\mathbf{X_B}^\intercal \mathbf{X_B})$ will not have inverse.

- We can use pseudo-inverses. In `R` we use `ginv` de la libraria MASS

- In other cases, we may have closed formulas for the coefficients: for example, if we have $X_i \in \mathbb{R}$ and $Y_i = \beta_0 + \beta_1 X_i$ then

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2}, \quad \text{and} \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

  where the application of bootstrap is much easier.

Bootstrap for Regression: A semiparametric Bootstrap

Consider the regression model

$$Y_i = \beta_0 + \beta^\intercal X_i + \varepsilon_i$$

with $Y_i \in \mathbb{R}$ and $X_i \in \mathbb{R}^d$, and suppose that ε_i are i.i.d. with mean 0 (which may be a strong assumption).

The idea of the semiparametric bootstrap is to notice that if the noise/error terms are i.i.d, then, we can permute them to obtain new pairs (Y_i^B, X_i). This is good because it is weird to have repeated observations in regression due to the inverse issues.

That is, we keep X_i fixed, but get a new value of Y_i^* by permuting the error terms.

Semiparametric Bootstrap for regression

1. Compute an estimate $\hat{\beta}_0$ and $\hat{\beta}$ of β_0 and β, respectively.
2. Compute $\hat{\varepsilon}_i = Y_i - (\hat{\beta}_0 - \hat{\beta}^\intercal X_i)$. Denote *hat*$\varepsilon$ the vector with all the $\hat{\varepsilon}_i$.
3. Let $(\sigma_j)_{j=1}^M$ be i.i.d. random permutations of $\{1, 2, \ldots, M\}$
4. For each j in $\{1, \ldots, M\}$, denote $\hat{\varepsilon}_{\sigma_j}$ as $\hat{\varepsilon}$ permuted by σ_j
5. For i in $\{1, \ldots, n\}$ and j in $\{1, \ldots, M\}$ do

$$Y_{ij}^B = \hat{\beta}_0 + \hat{\beta}^\intercal X_i + \hat{\varepsilon}_{\sigma_j}(i)$$